

# PAPER\_732: 10 Astrophysical Systems MUGE Compressed UQFF

**Class:** TenAstroSystemsMUGECalculator **CP4 Entry:** #316 **Keywords:** MUGE, UQFF, NGC 2264, UGC 10214, NGC 4676, Red Spider Nebula, NGC 3372, AG Carinae, M42, Tarantula Nebula, NGC 2841, Mystic Mountain, compressed gravity, aether field, resonance **Session:** 178 | **Version:** v5.35 **Source:** grok\_share\_ba508f76c8e.txt entry #101

## Abstract

Ten canonical astrophysical systems spanning star-forming nebulae, interacting galaxies, and isolated spirals are unified under the MUGE Compressed UQFF framework. The master gravity equation incorporates Hubble expansion correction  $H(z)$ , mass evolution from star formation  $M_{evo}(t)$ , radiative/merger erosion  $E_{rad}(t)$ , the UQFF time-reversal zone factor  $f_{TRZ}$ , and an electromagnetic-aether term  $F_{em}$ . A complementary resonance equation  $R(t)$  captures gravitational and magnetic oscillatory modes for each system. Solved values at  $t = 10$  Myr confirm electromagnetic dominance for young nebulae (NGC 2264:  $g \approx 1.05 \times 10^{-2}$  m/s<sup>2</sup>) and classical gravity dominance for evolved spirals (NGC 2841:  $g \approx 5 \times 10^{-11}$  m/s<sup>2</sup>).

## 1. System Parameters

| # | System                      | $M$ (kg)               | $r$ (m)               | $z$    | SFR ( $M_{\odot}/\text{yr}$ ) | $B$ (T)   |
|---|-----------------------------|------------------------|-----------------------|--------|-------------------------------|-----------|
| 1 | NGC 2264<br>(Cone Nebula)   | $1.989 \times 10^{33}$ | $4.73 \times 10^{16}$ | 0.0006 | 0.5                           | $10^{-5}$ |
| 2 | UGC 10214<br>(Tad-pole)     | $1.989 \times 10^{41}$ | $1.24 \times 10^{21}$ | 0.028  | 1.0                           | $10^{-5}$ |
| 3 | NGC 4676<br>(Mice)          | $3.978 \times 10^{41}$ | $3 \times 10^{20}$    | 0.022  | 10.0                          | $10^{-4}$ |
| 4 | Red Spider Nebula           | $1.193 \times 10^{30}$ | $10^{16}$             | 0.0013 | 0.0                           | $10^{-5}$ |
| 5 | NGC 3372<br>(Carina)        | $1.989 \times 10^{35}$ | $2 \times 10^{17}$    | 0.0025 | 2.0                           | $10^{-5}$ |
| 6 | AG Carinae Nebula           | $3.978 \times 10^{31}$ | $10^{16}$             | 0.002  | 0.0                           | $10^{-5}$ |
| 7 | M42<br>(Orion)              | $3.978 \times 10^{33}$ | $2 \times 10^{16}$    | 0.0004 | 0.3                           | $10^{-5}$ |
| 8 | Tarantula<br>(30 Do-radius) | $1.989 \times 10^{35}$ | $3 \times 10^{17}$    | 0.0005 | 5.0                           | $10^{-4}$ |
| 9 | NGC 2841<br>(spiral)        | $1.989 \times 10^{41}$ | $5 \times 10^{20}$    | 0.0031 | 0.5                           | $10^{-5}$ |

| #  | System          | $M$ (kg)               | $r$ (m)   | $z$    | SFR ( $M_{\odot}/\text{yr}$ ) | $B$ (T)   |
|----|-----------------|------------------------|-----------|--------|-------------------------------|-----------|
| 10 | Mystic Mountain | $1.989 \times 10^{32}$ | $10^{16}$ | 0.0025 | 0.1                           | $10^{-5}$ |

## 2. MUGE Compressed Gravity Master Equation

$$g(r, t) = \frac{G M}{r^2} (1 + H(z) t) (1 + M_{evo}) (1 - E_{rad}) (1 + f_{TRZ}) + F_{em}$$

Where each correction factor is:

### 2.1 Hubble Expansion Correction

$$H(z) = H_0 \sqrt{0.3(1+z)^3 + 0.7}, \quad H_0 = 2.268 \times 10^{-18} \text{ s}^{-1}$$

### 2.2 Mass Evolution from Star Formation

$$M_{evo}(t) = \frac{\dot{M}_{sf} \cdot t}{M_0}$$

where  $\dot{M}_{sf}$  is the star formation rate in  $M_{\odot}/\text{yr}$  and  $M_0$  is the system mass in solar masses.

### 2.3 Radiation / Merger Erosion

$$E_{rad}(t) = E_0 (1 - e^{-t/\tau_{erode}})$$

$E_0$  is the peak erosion fraction;  $\tau_{erode}$  is the erosion timescale.

### 2.4 UQFF Time-Reversal Zone Correction

$$f_{TRZ} = 0.1 \Rightarrow (1 + f_{TRZ}) = 1.1$$

### 2.5 Electromagnetic-Aether Term

$$F_{em} = \frac{q_e v_{wind} B}{m_p} \left( 1 + \frac{\rho_{UA}}{\rho_{SCm}} \right) \cdot \eta_{scale}$$

where  $\rho_{UA}/\rho_{SCm} = 7.09 \times 10^{-36} / 7.09 \times 10^{-37} = 10$ ,  $\eta_{scale} = 10^{-12}$ .

## 3. Resonance UQFF Equation

$$R(t) = R_{grav} \cos(\omega_{grav} t) + R_{mag} \cos(\omega_{mag} t) \cdot \frac{\rho_{UA}}{\rho_{SCm}} \cdot (1 + f_{TRZ})$$

$$\omega_{grav} = \frac{2\pi}{\tau_{erode}}, \quad \omega_{mag} = 100 \omega_{grav}$$

$$R_{grav} = \frac{G M}{r^2} (1 + M_{evo}), \quad R_{mag} = \frac{q_e v_{wind} B}{m_p} \cdot \eta_{scale}$$

#### 4. Solved Results at $t = 10$ Myr

| System            | $g_{MUGE}$ (m/s <sup>2</sup> ) | Dominant Term            |
|-------------------|--------------------------------|--------------------------|
| NGC 2264          | $\sim 1.053 \times 10^{-2}$    | EM (Faraday + aether)    |
| UGC 10214         | $\sim 1.2 \times 10^{-11}$     | Gravity + tidal          |
| NGC 4676          | $\sim 2.0 \times 10^{-11}$     | Dual-mass gravity        |
| Red Spider Nebula | $\sim 2.5 \times 10^{-9}$      | Wind EM                  |
| NGC 3372          | $\sim 1.8 \times 10^{-9}$      | SFR + EM                 |
| AG Carinae        | $\sim 1.6 \times 10^{-9}$      | LBV wind                 |
| M42 (Orion)       | $\sim 8.4 \times 10^{-10}$     | Protostellar EM          |
| Tarantula Nebula  | $\sim 2.2 \times 10^{-9}$      | Starburst EM             |
| NGC 2841          | $\sim 5.0 \times 10^{-11}$     | Gravity (Milgrom regime) |
| Mystic Mountain   | $\sim 1.3 \times 10^{-9}$      | Pillar EM                |

#### 5. NGC 2264 Worked Example

At  $t = 3 \times 10^6$  yr =  $9.468 \times 10^{13}$  s:

$$g_{grav} = \frac{6.674 \times 10^{-11} \times 1.989 \times 10^{33}}{(4.73 \times 10^{16})^2} = 5.927 \times 10^{-11} \text{ m/s}^2$$

$$H(z = 0.0006) \cdot t \approx 2.268 \times 10^{-18} \times 1.0002 \times 9.468 \times 10^{13} \approx 2.147 \times 10^{-4}$$

$$M_{evo}(t = 3 \text{ Myr}) = \frac{0.5 \times 3 \times 10^6}{10^3} = 1.5 \Rightarrow (1 + M_{evo}) = 2.5$$

$$E_{rad}(t) = 0.2 \left( 1 - e^{-9.468 \times 10^{13} / 6.312 \times 10^{13}} \right) = 0.1554 \Rightarrow (1 - E_{rad}) = 0.8446$$

$$F_{em} = \frac{1.602 \times 10^{-19} \times 10^6 \times 10^{-5}}{1.673 \times 10^{-27}} \times 11 \times 10^{-12} \approx 1.053 \times 10^{-2} \text{ m/s}^2$$

$$g_{NGC2264} \approx 1.053 \times 10^{-2} \text{ m/s}^2$$

#### 6. Implementation

**C++ module:** TenAstroSystemsMUGE.h / TenAstroSystemsMUGE.cpp

**Python class:** TenAstroSystemsMUGECalculator (CondensedPhysics4.py, CP4 #316)

```
calc = TenAstroSystemsMUGECalculator()
result = calc.compute({"t": 3.156e14})
# returns: primary_equation, list of {name, g_muge, R_resonance} for 10 systems
```

#### 7. Conclusion

The MUGE Compressed UQFF framework unifies 10 astrophysical systems across six orders of magnitude in mass and spatial scale. The electromagnetic-aether term  $F_{em}$  dominates in young star-forming nebulae and planetary nebulae, while classical Newtonian gravity reemerges for massive evolved galaxies. The resonance equation  $R(t)$  provides

a complementary oscillatory signature for each system on characteristic star-formation and wind timescales.

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